Question ID: 52e589f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$m(t) = -0.0274\left(\frac{t}{7}\right)^2 + 7.3873\left(\frac{t}{7}\right) + 75.032$$

The function m gives the predicted body mass $m(t)$, in $\text{kilograms}\ (\text{kg})$, of a certain animal t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement "$m(330)$ is approximately equal to 362" in this context?

Answer

A. The predicted body mass of the animal was approximately $330\ \text{kg}$ 362 days after it was born.

B. The predicted body mass of the animal was approximately $362\ \text{kg}$ 330 days after it was born.

C. The predicted body mass of the animal was approximately $362\ \text{kg}$ $\frac{330}{7}$ days after it was born.

D. The predicted body mass of the animal was approximately $\frac{330}{7}\ \text{kg}$ 362 days after it was born.

Correct Answer: B

Rationale

Choice B is correct. In the statement "$m(330)$ is approximately equal to 362," the input of the function, 330, is the value of t, the elapsed time, in days, since the animal was born. The approximate value of the function, 362, is the predicted body mass, in kilograms, of the animal after that time has elapsed. Therefore, the predicted body mass of the animal was approximately $362\ \text{kg}$ 330 days after it was born.

Choice A is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to 330."

Choice C is incorrect. The number $\frac{330}{7}$ is the number of weeks, not the number of days, after the animal was born.

Choice D is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to $\frac{330}{7}$."